Probability Theory

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These are my notes on probability theory, which I am studying for C271A and as an extension of what I learned in analysis. I hope to cover the relevant measure theory whenever possible, and I will also add completed practice problems as well.

Textbooks Used to Develop these Notes:

1. ***Probability: Theory and Examples*** - *Rick Durrett*
2. ***Real Analysis*** - *Gerald Folland*

Other Textbooks Referenced:

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NOTE: Any sections/subsections marked with an asterisk are extra sections and have been added to this document as extensions of introduced theory.

1 Measure Spaces

1.1 Algebras and Rings

We begin with some basic measure theory which will motivate the formulation of the σ-algebra and allow us to go from generality to specificity in defining probability spaces.

We begin by defining a probability space as a triple $(\Omega, \mathcal{F}, P)$. The first element Ω is called the set of outcomes or the "sample space", which is just an underlying set that can be defined as we wish.

Definition 1.1 (σ-algebra)**.** *The second element $\mathcal{F}$ is called a σ-algebra, satisfying the following properties:*

1. $\Omega \in \mathcal{F}$
2. $\forall S, P \in \mathcal{F} : S \setminus P \in \mathcal{F}$
3. *The countable (finite or countably infinite) union of sets in $\mathcal{F}$ is in $\mathcal{F}$:*

$$\forall \{S_n\}_{n\in\mathbb{N}} \in \mathcal{F}^{\mathbb{N}} : \bigcup_{n\in\mathbb{N}} S_n \in \mathcal{F}$$

We can equivalently replace this condition with closure under disjoint unions.

We call the set S an event (specifically an event with an assigned probability, which we shall see later). We also define an algebra over sets $\mathcal{A}$, which weakens condition 3 to finite unions.

Consider the following example of a set that is an algebra but not a σ-algebra, quite similar to the countable/co-countable σ-algebra. Define:

$$\mathcal{A} \coloneqq \{A \subseteq X : A \text{ finite} \vee A^c \text{ finite}\}$$

where X is infinite.

Lemma 1.1. *We can derive other properties of sets in the σ-algebra, such as:*

1. $\varnothing \in \mathcal{F}$
2. $\forall \{S_n\}_{n\in\mathbb{N}} \in \mathcal{F}^{\mathbb{N}}$*:*

$$\bigcap_{n\in\mathbb{N}} S_n \in \mathcal{F}$$

3. $\forall S \in \mathcal{F} : \Omega \setminus S \in \mathcal{F}$

These properties also hold for algebras in the finite sense.

Proof. **σ-algebras**: For $\varnothing \in \mathcal{F}$, this follows from the fact that $\Omega \in \mathcal{F}$ and from property two, $\Omega \setminus \Omega = \varnothing \in \mathcal{F}$.

The second statement follows from DeMorgan's Laws. If $S_n \in \mathcal{F}$, then $S_n^c \in \mathcal{F}$, which tells us that:

$$\bigcup_{n\in\mathbb{N}} S_n^c = \left(\bigcap_{n\in\mathbb{N}} S_n\right)^c \in \mathcal{F}$$

Once again by the completement axiom, we see that:

$$\bigcap_{n\in\mathbb{N}} S_n \in \mathcal{F}$$

As for the third statement, we know that since $\Omega \in \mathcal{F}$, then we have that for all $S \in \mathcal{F}$, that $\Omega \setminus S \in \mathcal{F}$. □

Proof. **Algebras**: $\varnothing \in \mathcal{F}$ follows from property 2.

Since $\Omega \in \mathcal{F}$, we have that $\forall S \in \mathcal{F} : \Omega \setminus S \in \mathcal{F}$. So, for any finite collection $\{S_i\}_{i=0}^n$, we have that $S_i^c \in \mathcal{F}$:

$$\bigcup_{i=0}^n S_i^c \in \mathcal{F} \implies \left(\bigcup_{i=0}^n S_i^c\right)^c = \bigcap_{i=0}^n S_i \in \mathcal{F}$$

□

Lemma 1.2. *The intersection of a family of σ-algebras for a set Ω is a σ-algebra.*

Proof. Let us call the family $\{\mathcal{F}_\alpha : \alpha \in I\}$ such that we have valid measurable spaces $(\Omega, \mathcal{F}_\alpha)$, then:

$$\mathcal{F} \coloneqq \bigcap_{\alpha\in I} \mathcal{F}_\alpha$$

is a valid sigma algebra such that $(\Omega, \mathcal{F})$ is a measurable space. We see that:

$$\forall \alpha \in I : \Omega \in \mathcal{F}_\alpha \implies \Omega \in \bigcap_{\alpha\in I} \mathcal{F}_\alpha$$

so property 1 in Definition 1.1 is fulfilled. Now, for some S in the intersection:

$$S \in \bigcap_{\alpha\in I} \mathcal{F}_\alpha \iff \forall \alpha \in I : S \in \mathcal{F}_\alpha$$

$$\implies \forall \alpha \in I : S^c \in \mathcal{F}_\alpha \iff S^c \in \bigcap_{\alpha\in I} \mathcal{F}_\alpha$$

thus property 2 is fulfilled. Now consider some $\{S_n\}_{n\in\mathbb{N}} \in \mathcal{F}^{\mathbb{N}}$, then:

$$\forall n \in \mathbb{N} : S_n \in \mathcal{F} \implies \forall \alpha \in I : S_n \in \mathcal{F}_\alpha$$

$$\implies \forall \alpha \in I : \{S_n\}_{n\in\mathbb{N}} \in \mathcal{F}_\alpha^{\mathbb{N}} \implies \forall \alpha \in I : \bigcup_{n\in\mathbb{N}} S_n \in \mathcal{F}_\alpha$$

$$\implies \bigcup_{n\in\mathbb{N}} S_n \in \mathcal{F}$$

so we have verified property 3. This is quite similar to topologies, where the intersection of topologies is a topology, but the union is not necessarily a topology. □

Consider some sample space $\Omega \coloneqq \{1, 2, 3, 4\}$, then consider:

$$\mathcal{F}_1 \coloneqq \{\Omega, \varnothing, \{1\}, \{2, 3, 4\}\}$$

$$\mathcal{F}_2 \coloneqq \{\Omega, \varnothing, \{2\}, \{1, 3, 4\}\}$$

both of which are σ-algebras. If we take the intersection:

$$\mathcal{F}_3 \coloneqq \mathcal{F}_1 \cap \mathcal{F}_2 = \{\Omega, \varnothing\}$$

which is the "coarsest" or trivial σ-algebra. Now taking the union:

$$\mathcal{F}_1 \cup \mathcal{F}_2 = \{\Omega, \varnothing, \{1\}, \{2\}, \{2,3,4\}, \{1,3,4\}\}$$

but by Definition 1.1 property 3, we would need $\{1,2\}$ to belong in the union, thus the union is not necessarily a σ-algebra.

An example of a σ-algebra is the countable σ-algebra, defined as:

$$\mathcal{F} \coloneqq \{S \subseteq \Omega : S \text{ countable} \vee S^c \text{ countable}\}$$

Proof. We have that $\Omega \in \mathcal{F}$ since $\varnothing$ is trivially countable. Now consider some $S \in \mathcal{F}$, then if S is countable, we know that $(S^c)^c$ is countable, thus $S^c \in \mathcal{F}$. If S is not countable, then we know that S^c must be countable, so $S^c \in \mathcal{F}$.

Now take a countable union of sets in the sequence $\{S_n\}_{n\in\mathbb{N}} \in \mathcal{F}^{\mathbb{N}}$. If each S_n is countable, then it follows that the union is in $\mathcal{F}$ from the fact that countable union of countable sets is countable. If each S_n is not countable, then:

$$\left(\bigcup_{n\in\mathbb{N}} S_n\right)^c = \bigcap_{n\in\mathbb{N}} S_n^c \in \mathcal{F}$$

since S_n not countable and $S_n \in \mathcal{F}$ means S_n^c is countable. Assume some are countable and some are not, then for the S_i that are not countable, we have that S_i^c is countable. This forces the complement of the union to be countable. □

Definition 1.2. *Given a space Ω and some set $S \subseteq \mathcal{P}(\Omega)$, we define the minimal algebra generated by S as:*

$$\mathcal{F}(S) \coloneqq \bigcap_{S\subseteq\mathcal{F}} \mathcal{F}$$

which is an algebra as per Lemma 1.2.

Consider two subsets $P \subseteq S \subseteq \mathcal{P}(\Omega)$, then we know that:

$$P \subseteq S \subseteq \mathcal{F}(S) \implies \mathcal{F}(P) \subseteq \mathcal{F}(S)$$

Since $\mathcal{F}(S)$ is a σ-algebra containing P thus belongs in the intersection of the minimal σ-algebra containing P. We say the Borel σ-algebra on $\mathbb{R}$ is one that is generated by the set of all open intervals:

$$I \coloneqq \{(a,b) : (\forall a, b \in \mathbb{R} : a < b)\} \subseteq \mathcal{P}(\mathbb{R})$$

$$\mathcal{B} \coloneqq \mathcal{F}(I)$$

The motivation of defining $\mathcal{B}$ as a minimal algebra of an algebra generated by open intervals is to essentially "hope" that unmeasurable/pathological sets like the Vitali set are eliminated from question. $\mathcal{B}$ will contain open intervals, closed intervals, and countable unions and intersections of them (so Cantor type sets are also included).

The issue is, while the Borel σ-algebra gives us candidates for what sets are measurable, it does not give the existence of a measure in $\mathbb{R}$.

Consider the minimal σ-algebra of an already existing σ-algebra $\mathcal{G}$, with the minimal σ-algebra being $\mathcal{F}(\mathcal{G})$. Then:

$$\mathcal{F}(\mathcal{G}) \coloneqq \bigcap_{\mathcal{G} \subseteq \mathcal{F}} \mathcal{F}$$

By definition, we have that $\mathcal{G} \subseteq \mathcal{F}(\mathcal{G})$. However, since $\mathcal{G} \subseteq \mathcal{G}$, then we have that:

$$\mathcal{F}(\mathcal{G}) \subseteq \mathcal{G} \implies \mathcal{F}(\mathcal{G}) = \mathcal{G}$$

We introduce a few more concepts:

Definition 1.3. *A ring $\mathcal{R} \subseteq \mathcal{P}(\Omega)$ is an algebera without the condition that $\Omega \in \mathcal{R}$. A σ-ring $\mathcal{R}_\sigma$ is a σ-algebra without the same condition.*

Lemma 1.3. *Properties of rings:*

1. *Rings (and respectively σ-rings) are closed under (countable) intersections*

Proof. **Property One:** Consider a ring $\mathcal{R}$ and a finite collection of sets $\{S_i\}_{i=0}^n \in \mathcal{R}^{n+1}$. We prove by induction, consider the base case $n = 1$. Then:

$$S_1 \cap S_0 = S_1 \setminus (S_1 \setminus S_0)$$

Since $S_1, S_0 \in \mathcal{R}$, then $S_1 \setminus S_0 \in \mathcal{R}$ and applying the set difference property again, we obtain that the intersection is in the ring. Now assume this to be true for some $n \in \mathbb{N}$. Then for $n+1$:

$$\bigcap_{i=0}^{n+1} S_i = S_{n+1} \cap \bigcap_{i=0}^{n} S_i = S_{n+1} \setminus \left(S_{n+1} \setminus \bigcap_{i=0}^{n} S_i \right) \in \mathcal{R}$$

Now consider a σ-ring $\mathcal{R}_\sigma$ and a sequence of sets $\{S_n\}_{n \in \mathbb{N}} \in \mathcal{R}_\sigma^{\mathbb{N}}$. We know that:

$$\bigcup_{n \in \mathbb{N}} S_n \in \mathcal{R}_\sigma$$

Now by the finite set difference property, we know that $E \setminus S_n \in \mathcal{R}_\sigma$, so their countable union is in the ring too. Now:

$$E \setminus \bigcup_{n \in \mathbb{N}} (E \setminus S_n) = E \cap \left(\bigcup_{n \in \mathbb{N}} (E \setminus S_n) \right)^c = E \cap \bigcap_{n \in \mathbb{N}} (E \setminus S_n)^c = E \cap \bigcap_{n \in \mathbb{N}} S_n$$

If we set $E = S_0$, then:

$$S_0 \cap \bigcap_{n \in \mathbb{N}} S_n = \bigcap_{n \in \mathbb{N}} S_n \in \mathbb{R}_\sigma$$

by once again applying the set difference property to E setminus the countable union (in the case that $E = S_0$). □

1.1.1 Problems (Folland and Durrett)

Folland 1.2.1D: If $\mathcal{R}_\sigma$ is a σ-ring, then $\mathcal{F} \coloneqq \{E \subseteq X : (\forall F \in \mathcal{R}_\sigma : E \cap F \in \mathcal{R}_\sigma)\}$ is a σ-algebra:

Proof. First we observe that X is in $\mathcal{F}$ since for all $F \in \mathcal{R}_\sigma$, we have that $X \cap F = F \in \mathcal{R}_\sigma$. Now, consider any $E, G \in \mathcal{F}$ such that:

$$\forall F \in \mathcal{R}_\sigma : E \cap F \in \mathcal{R}_\sigma \wedge G \cap F \in \mathcal{R}_\sigma$$

Then by set difference properties of the σ-ring, we have that:

$$(E \cap F) \setminus (G \cap F) \in \mathcal{R}_\sigma \implies (E \cap F) \cap (G^c \cup F^c) = E \cap G^c \cap F = (E \setminus G) \cap F \in \mathcal{R}_\sigma$$

thus $E \setminus G \in \mathcal{F}$. Finally, we must prove closure under countable union. Take a sequence $\{E_n\}_{n\in\mathbb{N}} \in \mathcal{F}^{\mathbb{N}}$ such that:

$$\forall F \in \mathcal{R}_\sigma, \forall n \in \mathbb{N} : E_n \cap F \in \mathcal{R}_\sigma$$

$$F \cap \bigcup_{n\in\mathbb{N}} E_n = \bigcup_{n\in\mathbb{N}} (F \cap E_n)$$

Since $F \cap E_n \in \mathcal{R}_\sigma$ for each E_n, then by closure under countable union of the σ-ring, we have that the countable union of E_n is in $\mathcal{F}$. Thus $\mathcal{F}$ is a σ-algebra. Here we used a σ-ring to describe a σ-algebra, a similar technique to what is used in Caratheodory's Extension Theorem. □

Folland 1.2.5: If $\mathcal{M}$ is the σ-algebra generated by $E \subseteq \mathcal{P}(X)$, then $\mathcal{M}$ is the union of the σ-algebras generated by $\mathcal{S}$ as $\mathcal{S}$ ranges over all countable subsets of E. In other words, we want to show:

$$\mathcal{F}(E) = \bigcup_{\alpha\in I} \mathcal{F}(S_\alpha)$$

Proof. Since $E \subseteq \mathcal{F}(E)$, we have that:

$$\forall \alpha \in I : S_\alpha \subseteq E \implies \forall \alpha \in I : \mathcal{F}(S_\alpha) \subseteq \mathcal{F}(E) \implies \bigcup_{\alpha\in I} \mathcal{F}(S_\alpha) \subseteq \mathcal{F}(E)$$

For the reverse inclusion, we show that the union is a sigma algebra itself. The space X is trivially in the union. If we take some P in the union, then:

$$\exists \alpha \in I : P \in \mathcal{F}(S_\alpha) \implies P^c \in \mathcal{F}(S_\alpha) \implies P^c \in \bigcup_{\alpha\in I} \mathcal{F}(S_\alpha)$$

Now consider some sequence $\{P_n\}_{n\in\mathbb{N}}$ made from elements in the union. Then we have that:

$$\forall n \in \mathbb{N}, \exists \alpha \in I : P_n \in \mathcal{F}(S_\alpha)$$

Denote each of these α as α_n based on the corresponding set P_n. Then we have that:

$$\bigcup_{n\in\mathbb{N}} S_{\alpha_n}$$

is a countable union of countable subsets of E and thus is a countable subset of E, so there must exist some α' such that:

$$\forall n \in \mathbb{N} : S_{\alpha_n} \subseteq \bigcup_{n\in\mathbb{N}} S_{\alpha_n} = S_{\alpha'} \implies \mathcal{F}(S_{\alpha_n}) \subseteq \mathcal{F}(S_{\alpha'})$$

Thus there exists an $\alpha' \in I$ such that:

$$\forall n \in \mathbb{N} : P_n \in \mathcal{F}(S_{\alpha'}) \subseteq \bigcup_{\alpha\in I} \mathcal{F}(S_\alpha)$$

so the union is a valid σ-algebra. Now if we define $E \coloneqq \{E_\beta : \beta \in J\}$, then we know that:

$$\forall \beta \in J, \exists \alpha'' \in I : E_\beta \in S_{\alpha''}$$

where the countable subset of E is simply $\{E_\beta\}$ for that specific $\beta \in J$. Thus:

$$E \subseteq \bigcup_{\alpha \in I} S_\alpha \subseteq \bigcup_{\alpha \in I} \mathcal{F}(S_\alpha) \implies \mathcal{F}(E) \subseteq \bigcup_{\alpha \in I} \mathcal{F}(S_\alpha)$$

where we once again applied the subset property and the fact that the σ-algebra generated by a σ-algebra is itself. Then if we take some $C \in \mathcal{F}(E)$, we have that:

$$\exists \alpha \in I : C \in \mathcal{F}(S_\alpha)$$

So any set in the σ-algebra generated by a possibly large (uncountable), complex set can be in the simpler σ-algebra generated by a countable subset of the set. □

Durrett 1.1.1: Let $\Omega = \mathbb{R}$, $\mathcal{F}$ be the σ-algebra of countable/co-countable sets, and $P(A)$ be 0 for a countable set and 1 for a cocountable set. Show that $(\Omega, \mathcal{F}, P)$ is a probability space.

Proof. We proved the countable/co-countable set is a σ-algebra, but we do it once again, this time using the set difference definition. We know that $\Omega \in \mathcal{F}$, since $\varnothing$ is countable.

Now consider any two subsets $E, F \in \mathcal{F}$. We can have the following situations:

1. E, F are both countable
2. E, F both have countable complements
3. E is countable and F has a countable complement
4. E has a countable complement and F is countable

In the first case, we have that $E \setminus F = E \cap F^c \subseteq E$, making it countable. The second case is similar, where $E \setminus F = E \cap F^c \subseteq F^c$, making it countable. This applies to the third case as well. For the fourth case $E \setminus F = E \cap F^c$. Its complement is $E^c \cup F$ which is countable, thus $E \setminus F \in \mathcal{F}$.

We use the same proof for countable unions, showing that $\mathcal{F}$ is a σ-algebra. Now since $\mathbb{R}$ is uncountable, we have that $P(\mathbb{R}) = 1$.

To prove countable additivity, assume each S_n in the sequence is countable and pairwise disjoint. Then the countable union is countable, so:

$$P\left(\bigcup_{n \in \mathbb{N}} S_n\right) = 0 = \sum_{n \in \mathbb{N}} P(S_n) = \sum_{n \in \mathbb{N}} 0$$

Now assume S_i, S_j have a countable complement such that $S_i \cap S_j = \varnothing$. Then:

$$S_i^c \cup S_j^c = \mathbb{R}$$

which is a contradiction since $\mathbb{R}$ is uncountable. So we can only have one such S_i in a disjoint collection with a countable complement (which by uncountability of reals is itself uncountable). In this case:

$$P\left(\bigcup_{n \in \mathbb{N}} S_n\right) = P\left(\bigcup_{n \neq i} S_n \cup S_i\right) = 1 = \sum_{n \neq i} 0 + 1$$

Since the union is uncountable, thus proving that $(\Omega, \mathcal{F}, P)$ is a probability space. □

Durrett 1.1.2: Show that $\mathcal{F}(S_d) = \mathcal{B}^d$ where $\mathcal{B}$ is the Borel set of $\mathbb{R}$. Where S_d is defined as:

$$S_d \coloneqq \{(a_1, b_1) \times ... \times (a_d, b_d) : (\forall i \in 1, ..., d : a_i < b_i)\}$$

Durrett 1.1.3:

Durrett 1.1.4: Show that if $\mathcal{F}_0 \subseteq \mathcal{F}_1 \subseteq ...$ are σ-algebras, then their countable union is an algebra. Given an example to show that the union need not be a σ-algebra.

Proof. We first see that the underlying space X is trivially in the union. Similarly, if:

$$P \in \bigcup_{n \in \mathbb{N}} \mathcal{F}_n \implies \exists n \in \mathbb{N} : P \in \mathcal{F}_n \implies P^c \in \mathcal{F}_n$$

Thus it is closed under complements. Now we must prove closure under finite unions. Consider a finite collection of elements in the union $\{P_i\}_{i=0}^n$, then:

$$\forall i \in 0, ..., n, \exists k \in \mathbb{N} : P_i \in \mathcal{F}_k$$

Since all the σ-algebras are nested, take $k(n) \coloneqq \max\{k \in \mathbb{N} : (\exists i \in 0, ..., n : P_i \in \mathcal{F}_k)\}$. In which case the finite union belongs to the union of the $\mathcal{F}_n$.

Consider the space $\mathbb{R}^{\mathbb{N}}$, the space of real-valued sequences. Then we have the nested σ-algebras:

$$\mathcal{F}(\{\{e_1\}\}) \subseteq \mathcal{F}(\{\{e_1\}, \{e_2\}\}) \subseteq ...$$

Observe that each of the σ-algebras under closure of unions only generate a finite set $\{e_1, ..., e_n\}$ (for the $n - th$ such σ-algebra). If we take the countable union of elements from the union:

$$E \coloneqq \bigcup_{n \in \mathbb{N}} \{e_n\} = \{e_1, e_2, ...\}$$

then we get the set of all canonical basis vectors, which cannot be contained in any of the σ-algebras. □

Durrett 1.1.5:

1.2 Measure

The third and final element of the triple is P, which we call the probability measure. A measure is defined as a following:

Definition 1.4. *$(X, \mathcal{F})$ is a measurable space on which we can put a measure $\mu : \mathcal{F} \to [0, \infty]$ with the properties:*

1. $\forall S \in \mathcal{F} : \mu(S) \geqslant 0 \land \mu(\varnothing) = 0$

   Note that we could also equivalently replace the second part with $\mu(\varnothing) \in [0, \infty)$.

2. *Given some $\{S_n\}_{n\in\mathbb{N}} \in \mathcal{F}^n$, if:*

$$\forall i \neq j : S_i \cap S_j = \varnothing \implies \mu\left(\bigcup_{n\in\mathbb{N}} S_n\right) = \sum_{n\in\mathbb{N}} \mu(S_n)$$

This second property will be quite important, and we shall call it countable additivity. Countable additivity gives a reason as to why we wanted property 3 in Definition 1.1, where:

$$\forall \{S_n\}_{n\in\mathbb{N}} \in \mathcal{F}^{\mathbb{N}} : \bigcup_{n\in\mathbb{N}} S_n \in \mathcal{F}$$

If this were not true, then we could have had some sequence of S_n in the σ-algebra where their countably infinite union is not measurable, so property 2 in this definition would not always hold.

Lemma 1.4. *Given a measure μ on a σ-algebra $\mathcal{F}$:*

$$\forall A \subseteq B : \mu(B) < \infty \implies \mu(B \setminus A) = \mu(B) - \mu(A)$$

Proof. We can write the disjoint union $B = (B \setminus A) \cup A$, so that (using countable additivity and empty sets to get finite additivity):

$$\mu(B) = \mu(B \setminus A) + \mu(A) \implies \mu(B \setminus A) = \mu(B) - \mu(A)$$

□

Lemma 1.5. *This definition of a measure reveals additional properties:*

1. *Monotonicity:*

$$\forall A, B \in \mathcal{F} : A \subseteq B \implies \mu(A) \leqslant \mu(B)$$

2. *Subadditivity:*

$$\forall \{S_n\}_{n\in\mathbb{N}} \in \mathcal{F}^{\mathbb{N}} : \mu\left(\bigcup_{n\in\mathbb{N}} S_n\right) \leqslant \sum_{n\in\mathbb{N}} \mu(S_n)$$

3. *Continuity from below:*

$$\forall n \in \mathbb{N} : S_n \subseteq S_{n+1} \implies \mu\left(\bigcup_{n\in\mathbb{N}} S_n\right) = \lim_{n\to\infty} \mu(S_n)$$

4. Continuity from above:

$$\forall n \in \mathbb{N} : S_{n+1} \subseteq S_n \wedge \mu(S_0) < \infty \implies \mu\left(\bigcap_{n\in\mathbb{N}} S_n\right) = \lim_{n\to\infty} \mu(S_n)$$

Monotonicity and Subadditivity. Given $A \subseteq B$, consider the disjoint union $(B \setminus A) \cup A$. Then countable addiivity and $\mu(\varnothing) = 0$, gives us that:

$$\mu(B) = \mu((B \setminus A) \cup A) = \mu(B \setminus A) + \mu(A) \geqslant \mu(A)$$

by non-negativity.

Now consider the countable union of subsets S_n. We can rewrite the union as a disjoint union:

$$\bigcup_{n\in\mathbb{N}} S_n = \bigcup_{n\in\mathbb{N}} \left(S_n \setminus \bigcup_{i=0}^{n-1} S_i\right)$$

$$\mu\left(\bigcup_{n\in\mathbb{N}} S_n\right) = \sum_{n\in\mathbb{N}} \mu\left(S_n \setminus \bigcup_{i=0}^{n-1} S_i\right) \leqslant \sum_{n\in\mathbb{N}} \mu(S_n)$$

by monotonicity. □

Continuity From Below. Now observe the decomposition from the previous step in the subadditivity proof:

$$\sum_{n\in\mathbb{N}} \mu\left(S_n \setminus \bigcup_{i=0}^{n-1} S_i\right) = \lim_{n\to\infty} \sum_{j=0}^{n} \mu\left(S_j \setminus \bigcup_{i=0}^{j-1} S_i\right)$$

Now the finite sum can be recombined by finite additivity since each subset is disjoint, obtaining:

$$= \lim_{n\to\infty} \mu\left(\bigcup_{j=0}^{n} S_j\right) = \mu\left(\bigcup_{n\in\mathbb{N}} S_n\right)$$

Continuity from below is a simple application of the fact that the subsets make non-decreasing chain. The finite intersection of those S_j subsets is simply S_n, thus:

$$\lim_{n\to\infty} \mu(S_n) = \mu\left(\bigcup_{n\in\mathbb{N}} S_n\right)$$

□

Continuity From Above, Proof 1. We offer two proofs, one using continuity from below and the other using countable additivity. Now consider a non-increasing chain of subsets with $\mu(A_0) < \infty$, then we have that:

$$A_0 \setminus A_0 \subseteq A_0 \setminus A_1 \subseteq A_0 \setminus A_2 \subseteq ...$$

since $A_0 \supseteq A_1 \supseteq ...$ tells us that $A_0^c \subseteq A_1^c \subseteq ....$ So applying continuity from above, we have that:

$$\lim_{n\to\infty} \mu(A_0 \setminus A_n) = \mu\left(\bigcup_{n\in\mathbb{N}} (A_0 \cap A_n^c)\right) = \mu\left(A_0 \setminus \bigcap_{n\in\mathbb{N}} A_n\right)$$

Now applying finite additivity to both sides, we obtain:

$$\mu(A_0 \setminus A_n) + \mu(A_n) = \mu(A_0) \wedge \mu\left(A_0 \setminus \bigcap_{n\in\mathbb{N}} A_n\right) + \mu\left(\bigcap_{n\in\mathbb{N}} A_n\right) = \mu(A_0)$$

Now $\mu(A_0) < \infty$ tells us that by monotonicity, $\mu(A_n) < \infty$, so we can subtract on both sides and rewrite our expression as:

$$\lim_{n\to\infty}(\mu(A_0) - \mu(A_n)) = \mu(A_0) - \mu\left(\bigcap_{n\in\mathbb{N}} A_n\right)$$

Thus:

$$\lim_{n\to\infty}\mu(A_n) = \mu\left(\bigcap_{n\in\mathbb{N}} A_n\right)$$

Observe that we dont necessarily require $\mu(A_0) < \infty$. We can have some $m \in \mathbb{N}$ such that $\mu(A_m) < \infty$, which gives us the result:

$$\lim_{n\to\infty}\mu(A_n) = \mu\left(\bigcap_{n\geqslant m} A_n\right)$$

Now if we take some x in the intersection, then:

$$\forall n \geqslant m : x \in A_n \implies x \in A_m \implies \forall j < m : x \in A_n$$

by the non-increasing property (the reverse inclusion follows from subset properties). Thus:

$$\lim_{n\to\infty}\mu(A_n) = \mu\left(\bigcap_{n\geqslant m} A_n\right) = \mu\left(\bigcap_{n\in\mathbb{N}} A_n\right)$$

□

Continuity From Above, Proof 2. While it was simple to use continuity from below, it is not required to prove continuity from above. Consider the disjoint union:

$$\bigcup_{n\in\mathbb{N}} A_n \setminus A_{n+1} = \bigcup_{n\in\mathbb{N}} A_n \cap \bigcup_{n\in\mathbb{N}} A_{n+1}^c = A_0 \setminus \bigcap_{n\in\mathbb{N}} A_{n+1}$$

Then from the previous proof:

$$\bigcap_{n\in\mathbb{N}} A_n = \bigcap_{n\in\mathbb{N}} A_{n+1}$$

Now applying countable additivity:

$$\mu\left(\bigcup_{n\in\mathbb{N}} A_n \setminus A_{n+1}\right) = \sum_{n\in\mathbb{N}} \mu(A_n \setminus A_{n+1}) = \lim_{n\to\infty}\sum_{i=0}^{n} \mu(A_i \setminus A_{i+1})$$

$$= \lim_{n\to\infty}\mu(A_0 \setminus A_{n+1})$$

Applying finite additivity on both sides, we get that:

$$\mu\left(\bigcup_{n\in\mathbb{N}} A_n \setminus A_{n+1}\right) = \mu\left(A_0 \setminus \bigcap_{n\in\mathbb{N}} A_n\right) = \mu(A_0) - \mu\left(\bigcap_{n\in\mathbb{N}} A_n\right)$$

Now the proof is the same as the previous one. We can also consider the more general case where $\mu(A_m) < \infty$ and then construct the disjoint union for $n \geqslant m$. □

The importance of highlighting these two proofs is to draw some important conclusions about sufficient conditions. If we were instead working with a premeasure μ_0 on an algebra, then the sufficient condition is that the countable union must belong to the algebra, $\mathcal{A}$, for continuity from below. For continuity from above, one sufficient condition we want is that there exists some $m \in \mathbb{N}$ such that $\mu(A_m) < \infty$. However, we still cannot say:

$$\bigcup_{n \geqslant m} A_n \setminus A_{n+1} = A_m \setminus \bigcap_{n \geqslant m} A_{n+1} \in \mathcal{A}$$

So, if we add the condition that:

$$\bigcap_{n \geqslant m} A_{n+1} \in \mathcal{A}$$

then since $\{A_n\}_{n \in \mathbb{N}} \in \mathcal{A}^{\mathbb{N}}$, we have that finite set differences belong in the algebra, thus:

$$A_m \setminus \bigcap_{n \geqslant m} A_{n+1} \in \mathcal{A}$$

Furthermore, we can provide an equivalent definition for measure which does not require countable additivity:

Lemma 1.6. *The function μ is a measure if and only if it is non-negative (with measure of $\varnothing$ finite), finitely additive, and continuous from below.*

Proof. $\implies$: This direction was proved in Lemma 1.3.

$\impliedby$: Consider a sequence of pairwise disjoint sets $A_1, A_2, ...$ Then we have the sequence:

$$\left\{ \bigcup_{i=0}^{n} A_n \right\}_{n \in \mathbb{N}}$$

which is non-decreasing (satisfying our conditiion for continuity from below). Then:

$$\mu\left(\bigcup_{n \in \mathbb{N}} A_n \right) = \mu\left(\bigcup_{n \in \mathbb{N}} \left(\bigcup_{i=0}^{n} A_n \right) \right) = \lim_{n \to \infty} \mu\left(\bigcup_{i=0}^{n} A_n \right)$$

By finite additivity, because the sequence is pairwise disjoint, we have that:

$$= \lim_{n \to \infty} \sum_{i=0}^{n} \mu(A_i) = \sum_{n \in \mathbb{N}} \mu(A_n)$$

Now we need to prove $\mu(\varnothing) = 0$. By finite additivity and $\mu(\varnothing) < \infty$:

$$\mu(\varnothing \cup \varnothing) = \mu(\varnothing) = \mu(\varnothing) + \mu(\varnothing) \implies \mu(\varnothing) = 0$$

So we must assume non-negativity (if we try to prove it via monotonicity, it becomes circular). Then we have that:

$$(\mu(\varnothing) \in [0, \infty)) + \text{finitely additive} + \text{continuous from below} \iff (\mu(\varnothing) = 0) + \text{countably additive}$$

$$\iff (\mu(\varnothing) \in [0, \infty)) + \text{countably additive}$$

□

Another useful lemma is the following:

Lemma 1.7. *A measure (or premeasure) is countably additive if and only if it is finitely additive and countably subadditive.*

Proof. $\implies$: Lemma 1.5 already proved this direction.

$\impliedby$: Finite additivity provides us with monotonicity, thus, consider a disjoint sequence of sets in the algebra/σ-algebra:

$$\mu\left(\bigcup_{n\in\mathbb{N}} A_n\right) \geqslant \mu\left(\bigcup_{i=0}^{n} A_i\right) = \sum_{i=0}^{n} \mu(A_i)$$

Since this is true for all $n \in \mathbb{N}$, we can take the limit on the right hand side and combined with countable subadditivity, we obtain:

$$\lim_{n\to\infty} \sum_{i=0}^{n} A_i = \sum_{n\in\mathbb{N}} A_n \leqslant \mu\left(\bigcup_{n\in\mathbb{N}} A_n\right) \leqslant \sum_{n\in\mathbb{N}} A_n$$

□

We can now make this notion more specific to probability and say:

Definition 1.5. *A probability measure is a function $P : \mathcal{F} \to [0, \infty]$ where:*

1. $P(\Omega) = 1$
2. *Given some sequence $\{S_n\}_{n\in\mathbb{N}} \in \mathcal{F}^{\mathbb{N}}$:*

$$\forall i \neq j : S_i \cap S_j = \varnothing \implies P\left(\bigcup_{n\in\mathbb{N}} S_n\right) = \sum_{n\in\mathbb{N}} P(S_n)$$

Lemma 1.8. *P is a measure on the measurable space $(\Omega, \mathcal{F})$ and:*

$$\forall S \in \mathcal{F} : P(\Omega \setminus S) = 1 - P(S)$$

Proof. Non-negativity and countable additivity follows by definition. We only need to prove that $\mu(\varnothing) = 0$. Simply take the sequence:

$$\Omega, \varnothing, \varnothing, ...$$

Then by countable additivity:

$$P(\Omega) = P(\Omega) + \sum_{n\in\mathbb{N}} P(\varnothing)$$

Since $P(\Omega) = 1$, we can apply cancellation, which forces $P(\varnothing) = 0$.

This is a sufficient condition for measure spaces in general. Given a non-negative function μ with countable additivity:

$$\exists S \in \mathcal{F} : (\mu(S) < \infty) + \text{countable additivity} \implies \mu(\varnothing) = 0$$

proving that μ is a measure on that measurable space.

We prove the second part now that we know that $P(\varnothing) = 0$. Because of this fact, we can say that countable additivity implies finite additivity, thus:

$$P(\Omega) = P(\Omega \setminus S \cup S) = P(\Omega \setminus S) + P(S)$$

$$\implies P(\Omega \setminus S) = 1 - P(S)$$

To do this proof, we require that $S \subseteq \Omega$ for the decomposition to hold. □

1.2.1 Problems

The Dirac Measure: Let X be the underling set and $\mathcal{M} = P(X)$ and some function $f : X \to [0, \infty]$, then we can define a measure:

$$\mu(E) \coloneqq \sum_{x \in E} f(x)$$

We then define the dirac meausre $\delta_{x_0}(E)$ by the fact that $f(x_0) = 1$ and $f(x) = 0$ for all $x \neq x_0$. In this case:

$$\forall E \in \mathcal{M} : \delta_{x_0}(E) \coloneqq \begin{cases} 0, & x_0 \notin E \\ 1, & x_0 \in E \end{cases}$$

We can easily check the null set has 0 measure and countable additivity holds (by the splitting of the sum for disjoint sets).

Folland 1.3.9: Given $(X, \mathcal{M}, \mu)$, then for $E, F \in \mathcal{M}$:

$$\mu(E) + \mu(F) = \mu(E \cup F) + \mu(E \cap F)$$

Proof.

$$\mu(F) = \mu(F \setminus E \cup (E \cap F)) = \mu(F \setminus E) + \mu(E \cap F)$$
$$\mu(E) = \mu(E \setminus F \cup (E \cap F)) = \mu(E \setminus F) + \mu(E \cap F)$$

Thus:

$$\mu(E) + \mu(F) = \mu(E \cap F) + \mu(F \setminus E) + \mu(E \setminus F) + \mu(E \cap F) = \mu(E \cap F) + \mu(E \cup F)$$

by finite additivity as $E \setminus F, F \setminus E, E \cap F$ are disjoint and their union is equal to the union of E and F.

This is familiar to the property of probability, where:

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

□

1.3 Caratheodory Extension Theorem

We say that a null set $S \in \mathcal{F}$ is one which has measure 0. If we consider any subset $P \subseteq S$, then by monotonicity of the measure, we have that:

$$\mu(P) \leqslant \mu(S) = 0 \implies \mu(P) = 0$$

however, P may not be part of $\mathcal{F}$. We note an important property that the **COUNTABLE** union of null sets is a null set since:

$$\mu\left(\bigcup_{n\in\mathbb{N}} S_n\right) \leqslant \sum_{n\in\mathbb{N}} \mu(S_n) = 0$$

which follows from countably subadditivity of the measure (Lemma 1.5.2).

Definition 1.6. *A measure μ is complete on a measure space $(X, \mathcal{M}, \mu)$ if its associated σ-algebra contains all subsets of any null set and if the measure of those subsets is 0.*

We can extend the measure on the σ-algebra by the following:

Lemma 1.9. *Given a measure space $(X, \mathcal{F}, \mu)$, we say that $\overline{\mu}$ is a complete measure on $\overline{\mathcal{M}}$, where:*

$$\overline{\mathcal{M}} \coloneqq \{E \cup F : (E \in \mathcal{M} \wedge \exists N \in \mathcal{N} : F \subseteq N)\}$$

where $\mathcal{N}$ is the set of null sets in $\mathcal{M}$. We claim that $\overline{\mu}$ is a unique extension and that:

$$\overline{\mu}(E \cup F) = \mu(E)$$

$\overline{\mathcal{M}}$ is called the completion of $\mathcal{M}$.

Proof. We aim to show that $\overline{\mathcal{M}}$ is a σ-algebra, acting as the domain of $\overline{\mu}$. The space X is in the set.

Now consider some $E \cup F \in \overline{\mathcal{M}}$, we have that:

$$(E \cup F)^c = ((E \cup N) \cap (N^c \cup F))^c = (E \cup N)^c \cup (N^c \cup F)^c = (E \cup N)^c \cup (N \cap F^c)$$

$$= (E \cup N)^c \cup (N \setminus F)$$

Since $\mathcal{M}$ was a σ-algebra, we have that $(E \cup N)^c$ and $N \setminus F \subseteq N \in \mathcal{N}$, thus the complement belongs to $\overline{\mathcal{M}}$.

Now consider the countable union condition of some sequence $\{E_n \cup F_n\}$:

$$\bigcup_{n\in\mathbb{N}} (E_n \cup F_n) = \bigcup_{n\in\mathbb{N}} E_n \cup \bigcup_{n\in\mathbb{N}} F_n$$

Since $\mathcal{M}$ is a σ-algebra, we have that the union of all E_n belongs to it. Additionally by subadditivity of the measure:

$$\bigcup_{n\in\mathbb{N}} F_n \subseteq \bigcup_{n\in\mathbb{N}} N_n \wedge \mu\left(\bigcup_{n\in\mathbb{N}} N_n\right) \leqslant \sum_{n\in\mathbb{N}} \mu(N_n) = 0$$

Thus the union of null sets is a null set, so the union of F_n is a subset of a null set, which means the countable union of $E_n \cup F_n$ is the union of a set E in $\mathcal{M}$ and a set F in $\mathcal{N}$, proving closure under countable union.

We now need to show that $\overline{\mu} : \overline{\mathcal{M}} \to [0, \infty]$. First we show that it is a function such that:

$$E_1 \cup F_1 = E_2 \cup F_2 \implies \overline{\mu}(E_1 \cup F_1) = \overline{\mu}(E_2 \cup F_2) \iff \mu(E_1) = \mu(E_2)$$

Observe that:

$$E_1 \subseteq E_2 \cup F_2 \wedge E_2 \subseteq E_1 \cup F_1$$

which implies

$$\mu(E_1) \leqslant \mu(E_2) + \mu(N_2) = \mu(E_2)$$
$$\mu(E_2) \leqslant \mu(E_1) + \mu(N_1) = \mu(E_1)$$

We see that all subsets of null sets have measure zero as:

$$\forall F \subseteq N : \overline{\mu}(F) = \overline{\mu}(\varnothing \cup F) = \mu(\varnothing) = 0$$

It also trivially follows that the measure of $\varnothing$ under the complete measure is 0. Finally, for countable additivity, consider a sequence of $E_n \cup F_n$ pairwise disjoint. By the fact that $\overline{\mathcal{M}}$ is a σ-algebra:

$$\overline{\mu}\left(\bigcup_{n\in\mathbb{N}}(E_n \cup F_n)\right) = \overline{\mu}\left(\bigcup_{n\in\mathbb{N}} E_n \cup \bigcup_{n\in\mathbb{N}} F_n\right) = \mu\left(\bigcup_{n\in\mathbb{N}} E_n\right)$$

Since $\forall i \neq j : (E_i \cup F_i) \cap (E_j \cup F_j) = \varnothing$, we have that:

$$(E_i \cap E_j) \cup (E_i \cap F_j) \cup (E_j \cap F_i) \cup (F_i \cap F_j) = \varnothing$$

This tells us that all the intersections must be the empty set, so $E_i \cap E_j = \varnothing$, thus by countable additivity of μ:

$$\mu\left(\bigcup_{n\in\mathbb{N}} E_n\right) = \sum_{n\in\mathbb{N}} \mu(E_n) = \sum_{n\in\mathbb{N}} \overline{\mu}(E_n \cup F_n)$$

We finally need to show uniqueness. Assume there exists another such complete measure μ' such that:

$$\forall F \subseteq N : \mu'(F) = 0$$

so the two measures agree on subsets of null sets and on $\mathcal{M}$, in which case we want to show that:

$$\overline{\mu}(E \cup F) = \mu'(E \cup F)$$

We can write the following since $\mathcal{N} \subseteq \mathcal{M}$:

$$\mu'(E \cup F) \leqslant \mu'(E \cup N) = \overline{\mu}(E \cup N) \leqslant \overline{\mu}(E) \leqslant \overline{\mu}(E \cup F)$$

We can derive a similar inequality starting with $\overline{\mu}$, showing that the two agree on the completion of the σ-algebra. □

Definition 1.7. *Given a space X, we define an outer measure $\mu^* : \mathcal{P}(X) \to [0, \infty]$ with the following properties:*

1. $\mu^*(\varnothing) = 0$
2. $\forall A, B \in \mathcal{P}(X) : A \subseteq B \implies \mu^*(A) \leqslant \mu^*(B)$
3. $\forall \{A_n\}_{n\in\mathbb{N}} \in \mathcal{P}(X)^{\mathbb{N}} : \mu^*\left(\bigcup_{n\in\mathbb{N}} A_n\right) \leqslant \sum_{n\in\mathbb{N}} \mu^*(A_n)$

We require property two of monotonicity because we no longer have countable additivity to derive it. Instead we only have subadditivity.

We propose a specific formulation of the outer measure for some space X, where:

$$\forall A \in \mathcal{P}(X) : \mu^*(A) \coloneqq \inf\left\{\sum_{n\in\mathbb{N}} \mu_0(E_n) : A \subseteq \bigcup_{n\in\mathbb{N}} E_n\right\}$$

Right now, μ_0 is a blackbox function. It maps some $E \subseteq \mathcal{P}(X)$ to $[0, \infty]$ such that $\mu_0(\varnothing) = 0$ and $X \in E$. This means for any set A in the powerset, the outer measure is always defined since a covering sequence where $E_n = X$ always exists.

Additionally by the positivity of μ_0 there is always a lower bound for the set, thus the infimum exists. We prove that this formulation is in actuality an outer measure:

Proof. We see that $\mu^*(\varnothing) = 0$, simply consider the covering sequence $\varnothing, \varnothing, ...$

For property 2, monotonicity, observe that for all covering sequences of the set B:

$$A \subseteq B \subseteq \bigcup_{n\in\mathbb{N}} E_n \implies \mu^*(A) \leqslant \sum_{n\in\mathbb{N}} \mu_0(E_n)$$

By properties of infimum, we have that $\mu^*(A) \leqslant \mu^*(B)$ (greatest lower bound).

Now consider the sequence of subsets in the powerset $\{A_n\}_{n\in\mathbb{N}}$. Then:

$$\forall n \in \mathbb{N} : A_n \subseteq \bigcup_{m\in\mathbb{N}} E_m^{(n)}$$

Now consider some value $\epsilon/2^n$, for which by definition of infimum, there exists some covering sequence for A_n such that:

$$\sum_{m\in\mathbb{N}} \mu_0(E_m^{(n)}) \leqslant \mu^*(A_n) + \frac{\epsilon}{2^n}$$

Additionally, we have that:

$$\bigcup_{n\in\mathbb{N}} A_n \subseteq \bigcup_{n\in\mathbb{N}} \bigcup_{m\in\mathbb{N}} E_m^{(n)}$$

where the covering sequence is countable (this is because there is a bijection between $\mathbb{N}^2$ and $\mathbb{N}$). So by infimum properties, we have that:

$$\mu^*\left(\bigcup_{n\in\mathbb{N}} A_n\right) \leqslant \sum_{n\in\mathbb{N}} \sum_{m\in\mathbb{N}} \mu_0(E_m^{(n)}) \leqslant \sum_{n\in\mathbb{N}} \left(\mu^*(A_n) + \frac{\epsilon}{2^n}\right)$$

$$\leqslant \sum_{n\in\mathbb{N}} \mu^*(A_n) + 2\epsilon$$

Since this is true for all $\epsilon > 0$, we have proven countable subadditivity. Finite subadditivity follows from filling the remaining spots in the sequence of subsets with empty sets. □

Definition 1.8. *We say a set A is Caratheodory measurable or μ^*-measurable if:*

$$\forall E \in \mathcal{P}(X) : \mu^*(E) = \mu^*(E \cap A) + \mu^*(E \cap A^c)$$

In proving a set A is μ^*-measurable, it suffices to prove:

$$\mu^*(E) \geqslant \mu^*(E \cap A) + \mu^*(E \cap A^c)$$

As writing $E = E \cap (A \cup A^c) = (E \cap A) \cup (E \cap A^c)$, we obtain by subadditivity of the outer measure:

$$\mu^*(E) \leqslant \mu^*(E \cap A) + \mu^*(E \cap A^c)$$

Theorem 1.1 (Caratheodory's Extension Theorem)**.** *If μ^* is an outer measure on some space X, then:*

$$\mathcal{M} \coloneqq \{A \subseteq X : (\forall E \subseteq X : \mu^*(E) = \mu^*(E \cap A) + \mu^*(E \cap A^c))\}$$

is a complete σ-algebra, with $\mu^*|_{\mathcal{M}}$ being a complete measure.

Proof. We see that $X \in \mathcal{M}$ trivially. Additionally, we see that for any $A \in \mathcal{M}$, we have that $A^c \in \mathcal{M}$ by definition.

We now prove that $\mathcal{M}$ is closed under finite unions. Consider $A, B \in \mathcal{M}$. Then:

$$\mu^*(E \cap (A \cup B)) + \mu^*(E \cap A^c \cap B^c)$$

Simiar to Folland 1.3.9, we can write $A \cup B = (A \setminus B) \cup (A \cap B) \cup (B \setminus A)$, thus:

$$= \mu^*((E \cap A \cap B^c) \cup (E \cap A \cap B) \cup (E \cap B \cap A^c)) + \mu^*(E \cap A^c \cap B^c)$$

$$\leqslant \mu^*(E \cap A \cap B^c) + \mu^*(E \cap A \cap B) + \mu^*(E \cap B \cap A^c) + \mu^*(E \cap A^c \cap B^c)$$

Since A, B is μ^*-measurable:

$$= \mu^*(E \cap A) + \mu^*(E \cap A^c) = \mu^*(E)$$

Now that we have the property of finite unions, we can say $\mathcal{M}$ is an algebra. To upgrade it to a σ-algebra, we require closure under countable unions.

Now, we show that μ^* is finitely additive such that for some finite sequence of pairwise disjoint sets A_i in $\mathcal{M}$:

$$\mu^*\left(E \cap \bigcup_{i=0}^{n} A_i\right) = \sum_{i=0}^{n} \mu^*(E \cap A_i)$$

Consider the base case of $n = 1$, such that:

$$\mu^*(E \cap (A_0 \cup A_1)) = \mu^*((E \cap A_0 \cup E \cap A_1) \cap A_1) + \mu^*((E \cap A_0 \cup E \cap A_1) \cap A_1^c) = \mu^*(E \cap A_1) + \mu^*(E \cap A_0)$$

Now assume it holds true for some n, then for the $n+1$ case, since $A_{n+1} \in \mathcal{M}$:

$$\mu^*\left(\bigcup_{i=0}^{n+1} A_i\right) = \mu^*\left(\bigcup_{i=0}^{n} A_i \cup A_{n+1}\right)$$

$$= \mu^*\left(\left(\bigcup_{i=0}^{n} A_i \cup A_{n+1}\right) \cap A_{n+1}\right) + \mu^*\left(\left(\bigcup_{i=0}^{n} A_i \cup A_{n+1}\right) \cap A_{n+1}^c\right)$$

$$= \mu^*(A_{n+1}) + \mu^*\left(\bigcup_{i=0}^{n} A_i\right)$$

Nowe we can prove that $\mathcal{M}$ is closed under countable unions. Consider a sequence $\{A_n\}_{n\in\mathbb{N}} \in \mathcal{M}^{\mathbb{N}}$ and some $E \subseteq X$. By closure under finite unions:

$$\mu^*(E) = \mu^*\left(E \cap \bigcup_{i=0}^{n}\left(A_i \setminus \bigcup_{j=0}^{i-1} A_j\right)\right) + \mu^*\left(E \cap \bigcap_{i=0}^{n} A_i^c\right)$$

By finite additivity of the outer measure on the algebra $\mathcal{M}$, we have that:

$$\geqslant \sum_{i=0}^{n} \mu^*\left(E \cap A_i \setminus \bigcup_{j=0}^{i-1} A_j\right) + \mu^*\left(E \cap \bigcap_{n\in\mathbb{N}} A_n^c\right)$$

Since this is true for all $n \in \mathbb{N}$, we can take the limit as $n \to \infty$ and obtain:

$$\mu^*(E) \geqslant \sum_{n\in\mathbb{N}} \mu^* \left(E \cap A_n \setminus \bigcup_{i=0}^{n-1} A_i \right) + \mu^* \left(E \cap \bigcap_{n\in\mathbb{N}} A_n^c \right)$$

By subadditivity, we can recombine the sum and obtain:

$$\geqslant \mu^* \left(E \cap \bigcup_{n\in\mathbb{N}} A_n \right) + \mu^* \left(E \cap \bigcap_{n\in\mathbb{N}} A_n^c \right) \geqslant \mu^*(E)$$

We have shown an important equality, which we can state in full generality. Given some sequence of pairwise disjoint μ^*-measurable sets, we have that:

$$\mu^*(E) = \mu^* \left(E \cap \bigcup_{n\in\mathbb{N}} A_n \right) + \mu^* \left(E \cap \bigcap_{n\in\mathbb{N}} A_n^c \right) = \sum_{n\in\mathbb{N}} \mu^*(E \cap A_n) + \mu^* \left(E \cap \bigcap_{n\in\mathbb{N}} A_n^c \right)$$

Thus $\mathcal{M}$ is a σ-algebra. We now need to prove that μ^* is a measure on $\mathcal{M}$, for which it suffices to show countable additivity. Consider a disjoint sequence of sets $\{A_n\}_{n\in\mathbb{N}}$.

Then we have that their countable union is in the σ-algebra, so we can apply the measurability property:

$$\mu^* \left(\bigcup_{n\in\mathbb{N}} A_n \right) = \mu^* \left(\bigcup_{n\in\mathbb{N}} A_n \cap \bigcup_{n\in\mathbb{N}} A_n \right) + \mu^* \left(\bigcup_{n\in\mathbb{N}} A_n \cap \bigcap_{n\in\mathbb{N}} A_n^c \right)$$

$$= \sum_{n\in\mathbb{N}} \mu^*(A_n)$$

which follows from the highlighted equality.

Now we have that μ^* is a full measure on the σ-algebra $\mathcal{M}$. To show it is complete, consider a set E a subset of a null set N. Then taking any arbitrary $A \subseteq X$:

$$\mu^*(A) \leqslant \mu^*(A \cap E) + \mu^*(A \cap E^c) \leqslant \mu^*(N) + \mu^*(A \cap E^c) \leqslant \mu^*(A)$$

thus proving that $E \in \mathcal{M}$. By monotonicity of measure, we see that $\mu^*(E) = 0$. □

Definition 1.9 (Premeasure)**.** *We now introduce the concept of a premeasure $\mu_0 : \mathcal{A} \to [0, \infty]$ which is a non-negative function whose domain is an algebra such that:*

1. $\mu_0(\varnothing) = 0$
2. *For all sequences $\{A_n\}_{n\in\mathbb{N}} \in \mathcal{A}^{\mathbb{N}}$:*

$$\forall i \neq j : A_i \cap A_j = \varnothing \wedge \bigcup_{n\in\mathbb{N}} A_n \in \mathcal{A} \implies \mu_0 \left(\bigcup_{n\in\mathbb{N}} A_n \right) = \sum_{n\in\mathbb{N}} \mu_0(A_n)$$

Observe that we place the additional sufficiency that the countable union must be in the algebra $\mathcal{A}$. This is not necessary for measures since their domains are σ-algebras which are automatically closed under countable unions, unlike algebras which are only closed under finite unions (and thus intersections).

Lemma 1.10. *Given a space X and a premeasure $\mu_0 : \mathcal{A} \to [0, \infty]$, we have that:*

1. $\mu^*|_{\mathcal{A}} = \mu_0$
2. $\forall A \in \mathcal{A} : A$ *is* μ^**-measurable*

where the outer measure μ^* is formulated as the infimum of sums of premeasures of covering sequences (view Definition 1.8).

Proof. **Part One:** If we consider some $A \in \mathcal{A}$, then by definition, we see that:

$$\mu^*(A) \leqslant \mu_0(A)$$

as $A, \varnothing, \varnothing, ...$ is a covering sequence for A. Now for the reverse inequality, since $\mu^*(A)$ exists, we have a covering sequence for A, such that:

$$A \subseteq \bigcup_{n\in\mathbb{N}} B_n \implies A = A \cap \bigcup_{n\in\mathbb{N}} B_n \in \mathcal{A}$$

We can rewrite A as a disjoint union:

$$A = \bigcup_{n\in\mathbb{N}} \left(A \cap B_n \setminus \bigcup_{i=0}^{n-1} B_i \right) \in \mathcal{A}$$

So by countable additivity and montonicity, we have that:

$$\mu_0(A) = \sum_{n\in\mathbb{N}} \mu_0 \left(A \cap B_n \setminus \bigcup_{i=0}^{n-1} B_i \right) \leqslant \sum_{n\in\mathbb{N}} \mu_0(A \cap B_n) \leqslant \sum_{n\in\mathbb{N}} \mu_0(B_n)$$

By the infimum definition of the outer measure μ^*, we see that $\mu_0(A) \leqslant \mu^*(A)$.

Part Two: We now want to show that:

$$\forall A \in \mathcal{A}, \forall E \subseteq X : \mu^*(E) = \mu^*(E \cap A) + \mu^*(E \cap A^c)$$

We have that for all $\epsilon > 0$, there is some covering sequence of E such that:

$$\mu^*(E) + \epsilon \geqslant \sum_{n\in\mathbb{N}} \mu_0(B_n) = \sum_{n\in\mathbb{N}} \mu_0(B_n \cap (A \cup A^c))$$

$$= \sum_{n\in\mathbb{N}} \mu_0((B_n \cap A) \cup (B_n \cap A^c))$$

Now we see that $B_n \cap A \cap B_n \cap A^c = \varnothing$ for all $n \in \mathbb{N}$, so by finite additivity of the premeasure:

$$= \sum_{n\in\mathbb{N}} (\mu_0(B_n \cap A) + \mu_0(B_n \cap A^c))$$

Now since the sequence of B_n covered E, we have that the sequence of $B_n \cap A$ covers $E \cap A$. Similarly for $E \cap A^c$. Thus by definition of the outer measure:

$$\geqslant \mu^*(E \cap A) + \mu^*(E \cap A^c)$$

Since this is true for all $\epsilon > 0$, we have that:

$$\mu^*(E) \geqslant \mu^*(E \cap A) + \mu^*(E \cap A^c)$$

The reverse inequality follows from the finite subadditivity of the outer measure μ^*. □

Now since $\mathcal{A}$ is μ^*-measurable, we have that $\mathcal{A} \subseteq \mathcal{M}$ as defined in the Caratheodory Extension Theorem. Thus tells us that $\mathcal{F}(\mathcal{A}) \subseteq \mathcal{M}$.

As a result, we have that $\mu^*|_{\mathcal{F}(\mathcal{A})}$ is a measure (countable additivity follows from countable additivity on $\mathcal{M}$).

1.4 The Lebesgue-Stieltjes Measure

Definition 1.10 (Semialgebra)**.** *A semialgebra $\mathcal{S} \subseteq \mathcal{P}(X)$ has the properties:*

1. $X, \varnothing \in \mathcal{S}$
2. *Closure under finite intersection:*

$$\forall \{A_i\}_{i=0}^n \in \mathcal{S}^{n+1} : \bigcap_{i=0}^n A_i \in \mathcal{S}$$

3. *Complements as disjoint finite unions:*

$$\forall A \in \mathcal{S}, \exists \{B_i\}_{i=0}^n \in \mathcal{S}^{n+1}, \forall i \neq j : B_i \cap B_j = \varnothing \wedge A^c = \bigcup_{i=0}^n B_i$$

Lemma 1.11. *We can generate an algebra from a semialgebra $\mathcal{A}(\mathcal{S})$, where:*

$$\mathcal{A}(\mathcal{S}) \coloneqq \left\{ \bigcup_{i=0}^n A_i : (\{A_i\}_{i=0}^n \in \mathcal{S}^{n+1} \wedge \forall i \neq j : A_i \cap A_j = \varnothing) \right\}$$

Proof. We prove $\mathcal{A}(S)$ is an algebra. Since $X \in \mathcal{S}$, we have that $X \in \mathcal{A}(\mathcal{S})$.

We aim to show that the algebra is closed under complements. Consider some $C \in \mathcal{A}(\mathcal{S})$, then:

$$C = \bigcup_{i=0}^n A_i$$

where each of the A_i subsets are pairwise disjoint. Now, taking the complement:

$$C^c = \bigcap_{i=0}^n A_i^c = \bigcap_{i=0}^n \left(\bigcup_{j=0}^m B_j^{(i)} \right)$$

where the union is a disjoint union (follows from Property 3 of semialgebras). When we expand the expression, we get unions of finite intersections of these $B_j^{(i)}$ sets (which belong in $\mathcal{S}$ by Property 2). By the distributive property of intersections over unions, if you take the intersection of any of two of these finite intersections, we see they are disjoint, thus $C^c \in \mathcal{A}(\mathcal{S})$.

Now, we can show closure under finite intersections, where:

$$\bigcap_{i=0}^n C_i = \bigcap_{i=0}^n \left(\bigcup_{j=0}^m A_j^{(i)} \right)$$

Once again distributing the intersection over the union, we get a union of finite intersections (which belong in $\mathcal{S}$), which we can prove is disjoint in the same manner. Combined with complement, it is closed under finite union. □

Lemma 1.12. *We define a half open interval as an interval $(a, b]$ or (a, ∞) where $-\infty \leqslant a < b < \infty$. We call this the semialgebra of half open intervals or $\mathcal{I}_h$.*

Proof. We see that $\mathbb{R} \in \mathcal{I}_h$ as the interval $(-\infty, \infty)$. Now consider the intersection of:

$$(a, b] \cap (c, d]$$

where, without loss of generality, $a < c$ and $b < d$, then we get the intersection is a half interval $(c, b]$. If one of the half intervals were a ray in the form of (a, ∞) then it is trivial that the intersection is a half interval.

Finally, if we take the complement of a half interval $(a, b]$, we get $(-\infty, a] \cup (b, \infty)$ which is the disjoint union of half intervals, thus in the semialgebra. This property is trivial for a ray. □

Now we can generate an algebra of half open intervals, which we call $\mathcal{A}(\mathcal{I}_h)$. We can take this one step further and show that the generated sigma algebra is actually the borel sigma algebra. Since $\mathcal{I}_h \subseteq \mathcal{A}(\mathcal{I}_h)$, we have that $\mathcal{F}(\mathcal{I}_h) \subseteq \mathcal{F}(\mathcal{A}(\mathcal{I}_h))$.

For the other inclusion, we see that for all $\mathcal{F}$ such that $\mathcal{I}_h \subseteq \mathcal{F}$, we have that arbitrary countable unions of the elements in $\mathcal{I}_h$ are included in each of these σ-algebras. Thus:

$$\forall \mathcal{F} \subseteq \mathcal{P}(\mathbb{R}) : \mathcal{I}_h \subseteq \mathcal{F} \implies \mathcal{A}(\mathcal{I}_h) \subseteq \mathcal{F}$$

Thus $\mathcal{A}(\mathcal{I}_h) \subseteq \mathcal{F}(\mathcal{I}_h)$, which tells us that $\mathcal{F}(\mathcal{A}(\mathcal{I})_h) \subseteq \mathcal{F}(\mathcal{I}_h)$. Now we defined the Borel σ-algebra $\mathcal{B}$ as the one generated by open intervals. For any half open interval $(a, b]$:

$$\bigcap_{n \in \mathbb{N}} \left(a, b + \frac{1}{n+1} \right) = (a, b] \in \mathcal{B} \implies \mathcal{I}_h \subseteq \mathcal{B}$$

Thus $\mathcal{F}(\mathcal{I}_h) \subseteq \mathcal{B}$ (we used closure under countable intersections property of σ-algebras). This also holds for rays. Now consider any open interval (a, b). Then:

$$\bigcup_{n \in \mathbb{N}} \left(a, b - \frac{1}{n+1} \right] = (a, b) \in \mathcal{F}(\mathcal{I}_h)$$

which tells us that $\mathcal{F}(\mathcal{I}_h) \subseteq \mathcal{B}$, so we have that $\mathcal{F}(\mathcal{A}(\mathcal{I}_h)) = \mathcal{F}(\mathcal{I}_h) = \mathcal{B}$. The importance of this is the σ-algebra we will be working on and defining a measure on is still the Borel σ-algebra.

We now begin to construct the Lebesgue measure.

Lemma 1.13. *Given a semialgebra $\mathcal{S}$ and a set function, $l : \mathcal{S} \to [0, \infty]$, such that:*

1. *$l(\varnothing) = 0$*
2. *Finitely Additive:*

$$\forall \{A_i\}_{i=0}^n \in \mathcal{S}^{n+1}, \forall i \neq j : A_i \cap A_j = \varnothing \implies l\left(\bigcup_{i=0}^n A_i \right) = \sum_{i=0}^n l(A_i)$$

3. *Countably subadditive:*

$$\forall \{A_n\}_{n \in \mathbb{N}} \in \mathcal{S}^{\mathbb{N}} : \bigcup_{n \in \mathbb{N}} A_n \in \mathcal{S} \implies l\left(\bigcup_{n \in \mathbb{N}} A_n \right) \leqslant \sum_{n \in \mathbb{N}} l(A_n)$$

Then l extends to a premeasure μ_0 on $\mathcal{A}(\mathcal{S})$ such that:

$$\forall A \in \mathcal{S} : \mu(A) = l(A)$$

Proof. Recall lemma 1.7, which tells us equivalent conditions for a premeasure. We cannot immediately say these hold for a set function l, since the semialgebra is not necessarily closed under finite unions. However, an algebra is, so if we are able to upgrade the set function to the algebra, then we can show it is a premeasure.

We define our function on $\mathcal{A}(\mathcal{S})$ as:

$$\forall \{A_i\}_{i=0}^n \in \mathcal{S}^{n+1}, \forall k \neq j : A_k \cap A_j = \varnothing \implies \mu_0\left(\bigcup_{i=0}^n A_i\right) = \sum_{i=0}^n l(A_i)$$

We first must show that μ_0 is well defined. If we consider any set $A \in \mathcal{A}(\mathcal{S})$, then it is very much possible that:

$$A = \bigcup_{i=0}^n A_i = \bigcup_{j=0}^m B_j$$

where both are disjoint unions of elements from the semialgebra. Now observe that:

$$\forall j \in \{0, ..., m\} : B_j = B_j \cap \bigcup_{i=0}^n A_i = \bigcup_{i=0}^n (B_j \cap A_i)$$

which is a disjoint union. Thus:

$$\mu_0\left(\bigcup_{j=0}^m B_j\right) = \sum_{j=0}^m l(B_j) = \sum_{j=0}^m l\left(\bigcup_{i=0}^n (B_j \cap A_i)\right)$$

By the finite additivity of l, we have that:

$$= \sum_{j=0}^m \sum_{i=0}^n l(B_j \cap A_i) = \sum_{i=0}^n \sum_{j=0}^m l(A_i \cap B_j)$$

which we obtain by simply reorganizing the terms and the summations. Using finite additivity of l once again:

$$= \sum_{i=0}^n l\left(\bigcup_{j=0}^m (A_i \cap B_j)\right) = \sum_{i=0}^n l(A_i) = \mu_0\left(\bigcup_{i=0}^n A_i\right)$$

Now μ_0 is a function on the algebra. Consider two disjoint sets in the algebra, $A, B \in \mathcal{A}(\mathcal{S})$. Then:

$$\mu_0(A \cup B) = \mu_0\left(\bigcup_{i=0}^n A_i \cup \bigcup_{j=0}^m B_j\right)$$

where we have that each both unions are disjoint unions of elements in $\mathcal{S}$ by definition, and:

$$A \cap B = \bigcup_{i=0}^n A_i \cap \bigcup_{j=0}^m B_j = \bigcup_{i,j} (A_i \cap B_j) = \varnothing$$

This forces each of the A_i, B_j to be disjoint with eachother, so the union of the unions is a disjoint union. Applying the definition of μ_0:

$$\mu_0\left(\bigcup_{i=0}^n A_i \cup \bigcup_{j=0}^m B_j\right) = \sum_{i=0}^n l(A_j) + \sum_{j=0}^m l(B_j) = \mu_0(A) + \mu_0(B)$$

proving that μ_0 is finitely additive (the proof can be completed via induction).

Now we must prove countably subadditivity of μ_0. Consider some sequence $\{A_n\}_{n\in\mathbb{N}} \in \mathcal{A}(\mathcal{S})^{\mathbb{N}}$, such that:

$$\bigcup_{n\in\mathbb{N}} A_n \in \mathcal{A}(\mathcal{S})$$

then, by definition, there exists some disjoint union of $C_i \in \mathcal{S}$ such that:

$$\bigcup_{n\in\mathbb{N}} A_n = \bigcup_{i=0}^{m} C_i \implies \mu_0\left(\bigcup_{n\in\mathbb{N}} A_n\right) = \mu_0\left(\bigcup_{i=0}^{m} C_i\right)$$

$$= \sum_{i=0}^{m} l(C_i) = \sum_{i=0}^{m} l\left(C_i \cap \bigcup_{n\in\mathbb{N}} A_n\right) = \sum_{i=0}^{m} l\left(\bigcup_{n\in\mathbb{N}} (C_i \cap A_n)\right)$$

We have that the countable union belongs in $\mathcal{S}$ since it is simply equal to C_i. Thus applying countable subadditivity of l:

$$\leqslant \sum_{i=0}^{m}\sum_{n\in\mathbb{N}} l(C_i \cap A_n) = \sum_{n\in\mathbb{N}}\sum_{i=0}^{n} l(A_n \cap C_i)$$

which we obtained by rearranging the terms. Now with the following:

$$\forall i \neq j : (A_n \cap C_i) \cap (A_n \cap C_j) = (A_n \cap A_n) \cap (C_i \cap C_j) = A_n \cap \varnothing = \varnothing$$

we apply the definition of μ_0:

$$= \sum_{n\in\mathbb{N}} \mu_0\left(A_n \cap \bigcup_{i=0}^{m} C_i\right) = \sum_{n\in\mathbb{N}} \mu_0(A_n)$$

Now it follows from definition that:

$$\forall A \in \mathcal{S} : \mu_0(A) = l(A) \implies \mu_0(\varnothing) = l(\varnothing) = 0$$

We finish the argument by using lemma 1.7 to show that μ_0 is countably additive, showing it is a premeasure. □

We now have all the steps to (FINALLY!) construct the Lebesgue-Stieltjes measure. Consider the semialgebra $\mathcal{I}_h$ as per lemma 1.12. We show the following:

Lemma 1.14. *Given a non-decreasing, right continuous function $F : \mathbb{R} \to \mathbb{R}$, we can define a set function $l : \mathcal{I}_h \to [0, \infty]$ such that:*

$$\forall (a, b] \in \mathcal{I}_h : l(a, b] = F(b) - F(a) \wedge l(\varnothing) = 0$$

AND

$$\forall \{I_j\}_{j=0}^{n} \in \mathcal{I}_h^{n+1}, \forall u \neq v : I_u \cap I_v = \varnothing \implies l\left(\bigcup_{j=0}^{n} I_j\right) = \sum_{i=0}^{n} l(I_j)$$

We furthermore have that l is finitely additive and countably subadditive, allowing us to extend it to a premeasure μ_0 on $\mathcal{A}(\mathcal{S})$.

Proof. Because F is non-decreasing, we see that l is non-negative. Additionally since F is non-decreasing:

$$F(\infty) \coloneqq \lim_{x\to\infty} F(x) = \sup_{x\in\mathbb{R}} F(x)$$

and similarly defined by infimum for the limit as $x \to -\infty$. This allows us to define the value of rays $(-\infty, b], (b, \infty) \in \mathcal{I}_h$ as:

$$\forall (\infty, b], (a, \infty) \in \mathcal{I}_h : l(-\infty, b] = F(b) - F(-\infty) \wedge l(a, \infty) = F(\infty) - F(a)$$

(We do not have indeterminate $\infty - \infty$ situations since F is a map between the reals and not extended reals).

We are given that l is finitely additive, so we must prove that l is countably subadditive. Consider $\{I_n\}_{n \in \mathbb{N}} \in \mathcal{I}_h^{\mathbb{N}}$ such that:

$$\bigcup_{n \in \mathbb{N}} I_n \in \mathcal{I}_h \implies \bigcup_{n \in \mathbb{N}} I_h = (a, b]$$

We assume the union to be the half-interval $(a, b]$ (though the proof is easily extendable to rays). Now define each $I_n \coloneqq (a_n, b_n]$. Using right continuity of F, we have that:

$$\forall n \in \mathbb{N}, \forall \epsilon > 0, \exists b_n' > b_n : F(b_n') - F(b_n) < \frac{\epsilon}{2^n}$$

Similarly there exists some a' such that $F(a') - F(a) < \epsilon$. Now:

$$(a', b] \subseteq [a', b] \subseteq (a, b] \subseteq \bigcup_{n \in \mathbb{N}} (a_n, b_n] \subseteq \bigcup_{n \in \mathbb{N}} (a_n, b_n')$$

Since $[a', b]$ is compact, any open cover has a finite subcover, thus:

$$(a', b] \subseteq [a', b] \subseteq \bigcup_{i=0}^{m} (a_i, b_i') \subseteq \bigcup_{i=0}^{m} (a_i, b_i']$$

$$l(a', b] = l\left((a', b] \cap \bigcup_{i=0}^{m} (a_i, b_i']\right) = l\left(\bigcup_{i=0}^{m} ((a', b] \cap (a_i, b_i'])\right)$$

Now define $(c_i, d_i] \coloneqq (a', b] \cap (a_i', b_i']$ and by the properties of $\mathcal{I}_h$, we have that $(c_i, d_i] \in \mathcal{I}_h$. Now consider the ordered set $\{c_i, d_i\} \cup \{a, b\}$ (which has no repeats and the members of which we shall rename to k_j). Then:

$$\bigcup_{i=0}^{m} (c_i, d_i] = \bigcup_{j=0}^{p-1} (k_j, k_{j+1}]$$

where the latter is now a disjoint union. By construction, we have that there exists some $0 \leqslant j_i < p_i \leqslant p - 1$ such that:

$$\forall i \in \{0, ..., m\} : (c_i, d_i] = \bigcup_{j=j_i}^{p_i - 1} (k_j, k_{j+1}]$$

thus:

$$l\left(\bigcup_{i=0}^{m} (c_i, d_i]\right) = l\left(\bigcup_{j=0}^{p-1} (k_j, k_{j+1}]\right) = \sum_{j=0}^{p-1} l(k_j, k_{j+1}] \leqslant \sum_{i=0}^{m} \sum_{j=j_i}^{p_i - 1} l(k_j, k_{j+1}]$$

which once again by finite additivity and monotonicity (which we can prove from finite additivity):

$$= \sum_{i=0}^{m} l(c_i, d_i] \leqslant \sum_{i=0}^{m} l(a_i, b_i'] = \sum_{i=0}^{m} (F(b_i') - F(a_i))$$

Now we have the inequality:

$$F(b) - F(a) - \epsilon \leqslant F(b) - F(a') \leqslant \sum_{i=0}^{m} (F(b_i') - F(a_i)) \leqslant \sum_{i=0}^{m} \left(F(b_i) - F(a_i) + \frac{\epsilon}{2^i}\right)$$

$$F(b) - F(a) - \epsilon \leqslant \sum_{n\in\mathbb{N}} \left(F(b_n) - F(a_n) + + \frac{\epsilon}{2^n}\right) = \sum_{n\in\mathbb{N}} (F(b_n) - F(a_n)) + 2\epsilon$$

$$\forall \epsilon > 0 : F(b) - F(a) \leqslant \sum_{n\in\mathbb{N}} (F(b_n) - F(a_n)) + 3\epsilon$$

proving countable subadditivity for l. So by lemma 1.13, we observe that l extends to a premeasure μ_0 on $\mathcal{A}(\mathcal{I}_h)$. □

Constructing the measure now is just a matter of applying the theorems in the previous section on the Caratheodory Extension Theorem. We first define our outer measure using the newly constructed premeasure as:

$$\mu^*(A) \coloneqq \inf\left\{\sum_{n\in\mathbb{N}} \mu_0(I_n) : \{I_n\}_{n\in\mathbb{N}} \in \mathcal{A}(\mathcal{I}_h)^{\mathbb{N}} \wedge A \subseteq \bigcup_{n\in\mathbb{N}} I_n\right\}$$

Then by lemma 1.10, we have that $\mu^*|_{\mathcal{A}(\mathcal{I}_h)} = \mu_0$ and that $\mathcal{F}(\mathcal{A}(\mathcal{I}_h)) \subseteq \mathcal{M}$ (or in other words is μ^*-measurable). By lemma 1.12, we know that $\mathcal{F}(\mathcal{A}(\mathcal{I}_h)) = \mathcal{B}$ (the Borel σ-algebra). So we can easily see that $\mu^*|_{\mathcal{M}}$ being a measure means $\mu^*|_{\mathcal{B}}$ is a measure which we denote as μ_F or the Lebesgue-Stieltjes measure! Remember that theorem 1.1 gave us a complete measure on $\mathcal{M}$ but that is not necessarily true for $\mathcal{B}$ since it is a subset of $\mathcal{M}$.

1.5 Uniqueness Of the Lebesgue-Stieltjes Measure

2 Random Variables and Distributions